

BWB Geometry Parameters:

What Must Stay Fixed, What Can Vary, and How Each One Reaches C_L and C_D

Abstract

This report classifies every parameter in the BWB geometry/aerodynamics code into two groups: parameters that must be held within strict limits to preserve a recognizable Blended-Wing-Body shape, and parameters that are free to vary for efficiency tuning. For each parameter, the exact governing equation, its location in the code, its valid range, and its downstream effect on C_L , C_D , and L/D are given explicitly.

1 The Two Categories, at a Glance

	FIXED (structural identity)	VARIABLE (efficiency tuning)
What it controls	Whether the shape is recognizably a BWB at all	How efficient that BWB is (C_L , C_D , L/D)
Consequence of violating	Shape stops being a BWB (becomes a delta wing, flying wing, or degenerate/self-intersecting surface)	None – efficiency changes, but the shape stays a valid BWB
Parameters	Chord taper order, sweep order, span proportion, loading exponents n_u, n_l , surface ordering $y_u > y_l$	Surface amplitudes A_u, A_l , angle of attack α , span efficiency e , skin friction C_f

Table 1: The fundamental split used throughout this report.

Why this split exists mathematically: the FIXED parameters appear in the *geometry-construction* equations (Section 2) – `local_chord(y)`, `leading_edge_x(y)` – which define the aircraft’s outer mold line. The VARIABLE parameters appear *only* in the surface-shape equations $y_u(x)$, $y_l(x)$ (Section 3) and the physics equations (Section 4) – they can be freely swept without ever making `local_chord(y)` or `leading_edge_x(y)` produce a degenerate planform.

2 FIXED Parameters: the Planform Skeleton

2.1 Chord taper order – must shrink root to tip

Listing 1: local_chord(y) – planform chord distribution

```
1 def local_chord(y):
2     ay = np.abs(y)
3     inner = INNER_ROOT - (INNER_ROOT - INNER_TIP) * np.clip(ay / INNER_SPAN, 0, 1)
4     outer = OUTER_ROOT - (OUTER_ROOT - OUTER_TIP) * np.clip((ay - INNER_SPAN) / OUTER_SPAN
5         , 0, 1)
6     return np.where(ay <= INNER_SPAN, inner, outer)
```

Constraint: $INNER_ROOT > INNER_TIP \geq OUTER_ROOT > OUTER_TIP > 0$.

Typical values (paper-matched): $86 > 56 \geq 56 > 15$ mm.

Why fixed: this equation is a linear interpolation between four control points; if the ordering is violated (e.g. $OUTER_TIP > INNER_ROOT$), the chord distribution develops a reversal – the wingtip becomes fatter than the centerbody, which is not a BWB by definition.

2.2 Sweep order – must decrease outboard

Listing 2: leading_edge_x(y) – planform sweep

```
1 def leading_edge_x(y):
2     ay = np.abs(y)
3     le_j = INNER_SPAN * np.tan(np.radians(INNER_SWEEP))
4     inner_le = ay * np.tan(np.radians(INNER_SWEEP))
5     outer_le = le_j + (ay - INNER_SPAN) * np.tan(np.radians(OUTER_SWEEP))
6     return np.where(ay <= INNER_SPAN, inner_le, outer_le)
```

Constraint: $INNER_SWEEP \geq OUTER_SWEEP$, both within $[25^\circ, 55^\circ]$.

Typical values: 45° inner, 35° outer.

Why fixed: this is the defining "arrow" planform of a BWB (steep centerbody sweep blending to a shallower outer-wing sweep). Reversing the order produces a forward-swept or gull-shaped outline inconsistent with any real BWB design, including this paper's own Table 6.

2.3 Span proportion – centerbody must stay small

Constraint: $INNER_SPAN / (INNER_SPAN + OUTER_SPAN) \in [0.15, 0.30]$.

Typical value: $22 / (22 + 70) = 0.24$.

Why fixed: the centerbody represents the cabin/fuselage-replacing section. If it grows to a large fraction of total span, the aircraft no longer reads as "a fuselage blended into wings" – it becomes a generic flying wing with no distinguishable body section.

3 FIXED Parameters: the Surface-Shape Sanity Limits

3.1 Loading exponents – must stay single-bump

Listing 3: The two independent surface equations

```

1 def y_upper(xc, A, n):
2     return A * xc**n * (1 - xc)
3
4 def y_lower(xc, A, n):
5     return -A * xc**n * (1 - xc)

```

Constraint: $n_u, n_l \in [0.5, 2.0]$.

Why fixed: $x^n(1-x)$ is a single-hump function only in this range. Below 0.5 the curve develops a near-vertical spike at the leading edge (physically impossible surface); above 2.0 the peak flattens into the trailing edge, producing a shape with almost no usable camber and an unrealistic near-cusp leading edge.

3.2 Surface ordering – must not self-intersect

Constraint: $y_u(x) > y_l(x)$ for all $x \in [0, 1]$, i.e. $A_u \cdot x^{n_u}(1-x) > -A_l \cdot x^{n_l}(1-x)$, which holds automatically whenever $A_u, A_l > 0$ (guaranteed by construction, but must be checked if either amplitude is allowed to go negative).

Why fixed: if $y_u < y_l$ anywhere, the "thickness" $y_u - y_l$ becomes negative there – the surface passes through itself, which cannot represent a real solid body.

Fixed Parameter	Symbol	Constraint	Typical value
Inner root/tip chord	c_{ir}, c_{it}	$c_{ir} > c_{it}$	86, 56 mm
Outer root/tip chord	c_{or}, c_{ot}	$c_{it} \geq c_{or} > c_{ot}$	56, 15 mm
Inner/outer sweep	Λ_i, Λ_o	$\Lambda_i \geq \Lambda_o$, both 25–55°	45°, 35°
Span proportion	$b_i/(b_i + b_o)$	0.15–0.30	0.24
Loading exponents	n_u, n_l	0.5–2.0	1.0

Table 2: All FIXED parameters, their symbols, and safe constraints.

4 VARIABLE Parameters: the Efficiency Dials

4.1 Upper surface amplitude, A_u – the primary lift lever

Path to C_L : A_u appears inside $y_u(x) \rightarrow$ shifts the DERIVED camber line $y_c(x) = \frac{1}{2}(y_u + y_l) \rightarrow$ shifts $dy_c/dx \rightarrow$ shifts $\alpha_{L=0}$ (via the real thin-airfoil integral, Eq. 2) $\rightarrow$ shifts C_L (Eq. 3).

Safe range: $A_u \in [0.02, 0.20]$ (2%–20% of chord). **Verified effect:** raising A_u from 0.10 to 0.15 raised C_L from 0.033 to 0.053 (code run, this conversation).

4.2 Lower surface amplitude, A_l – the opposing lift lever

Path to C_L : same mechanism as A_u , but with opposite sign in the camber-line formula – increasing A_l pulls $y_c(x)$ downward, reducing (or reversing) camber.

Safe range: $A_l \in [0.02, 0.20]$. **Verified effect:** raising A_l from 0.06 to 0.12 dropped C_L from 0.033 to 0.008 (code run, this conversation).

4.3 Angle of attack, α – flight condition, not geometry

Safe range: $\alpha \in [-2^\circ, 12^\circ]$ (beyond this, thin-airfoil theory’s linear assumption breaks down; real stall is not modeled here). Enters C_L directly (Eq. 3).

4.4 Span efficiency e and skin friction C_f – drag-side dials

Safe ranges: $e \in [0.7, 0.95]$, $C_f \in [0.003, 0.006]$. Neither affects C_L or the drawn shape – both act only on C_D (Eq. 5).

5 The Full Equation Chain: Geometry $\rightarrow C_L, C_D$

5.1 Step 1 – Derive the camber line from your two surfaces

$$y_c(x) = \frac{y_u(x) + y_l(x)}{2}, \quad \frac{dy_c}{dx} \text{ computed numerically.} \quad (1)$$

5.2 Step 2 – The real thin-airfoil zero-lift angle

$$\alpha_{L=0} = -\frac{1}{\pi} \int_0^\pi \frac{dy_c}{dx} \Big|_{x=\frac{1}{2}(1-\cos\theta)} (\cos\theta - 1) d\theta. \quad (2)$$

Verified against textbook values: NACA 2412 $\rightarrow -2.08^\circ$ (textbook -2.0 to -2.1°); NACA 4412 $\rightarrow -4.15^\circ$ (textbook -4.15°), exact match.

5.3 Step 3 – Finite-wing lift coefficient

$$C_L = \frac{2\pi AR}{AR+2}(\alpha - \alpha_{L=0}), \quad AR = \frac{(2b_{half})^2}{2S}. \quad (3)$$

5.4 Step 4 – Wetted-area parasite drag (thickness now genuinely matters)

$$C_{D,0} = C_f \cdot \frac{A_{\text{wetted}}}{S}, \quad A_{\text{wetted}} = \oint \sqrt{1 + \left(\frac{dy_u}{dx}\right)^2} dx + \oint \sqrt{1 + \left(\frac{dy_l}{dx}\right)^2} dx. \quad (4)$$

5.5 Step 5 – Induced drag and total efficiency

$$C_{D,i} = \frac{C_L^2}{\pi e AR}, \quad C_D = C_{D,0} + C_{D,i}, \quad L/D = \frac{C_L}{C_D}. \quad (5)$$

6 Complete Parameter Reference Table

Parameter	Type	Range	Feeds	Code location
$c_{ir}, c_{it}, c_{or}, c_{ot}$	Fixed	taper order	S, AR	<code>local_chord()</code>
Λ_i, Λ_o	Fixed	25–55°, ordered	shape only	<code>leading_edge_x()</code>
b_i, b_o (spans)	Fixed	$b_i/(b_i+b_o) \in [.15, .3]$	S, AR	HALF_SPAN, area calc
n_u, n_l	Fixed	0.5–2.0	$\alpha_{L=0}$ (shape only)	<code>y_upper()</code> , <code>y_lower()</code>
A_u (inner/outer)	Variable	0.02–0.20	$C_L \uparrow, C_{D,0} \uparrow$	<code>y_upper()</code>
A_l (inner/outer)	Variable	0.02–0.20	$C_L \downarrow, C_{D,0} \uparrow$	<code>y_lower()</code>
α	Variable	–2° to 12°	C_L directly	<code>alpha_rad</code>
e	Variable	0.7–0.95	$C_{D,i}$ only	CD_i formula
C_f	Variable	0.003–0.006	$C_{D,0}$ only	CD0 formula

Table 3: Complete classification of every parameter in the code.

Practical workflow: lock the five FIXED-category parameter groups at values satisfying their listed constraints once, at the start of a design study – this guarantees every subsequent shape is a valid BWB. Then sweep only A_u, A_l, α, e, C_f within their listed ranges to search for maximum L/D (e.g. via the bisection search demonstrated earlier in this conversation, which converged to $CD_i = CD_0$ at the optimum – the classic max-L/D condition).

7 Conclusion

The BWB code cleanly separates two kinds of parameters. Chord taper order, sweep order, span proportion, and the loading exponents n_u, n_l define the aircraft’s *identity* as a blended wing body – they must stay within the constraints in Table 3, or the geometry degenerates into a non-BWB shape. Surface amplitudes A_u, A_l , angle of attack, span efficiency, and skin friction are *performance dials* – freely tunable within their listed ranges, feeding directly and traceably into C_L (Eq. 3) and C_D (Eq. 5) without ever risking the shape’s validity. This separation is what makes systematic efficiency optimization possible: search freely in the second group, never in the first.